

GeoAI-Based Multi-Temporal Agricultural Change Detection and NDVI Trend Analysis

*Using Sentinel-2 in Chamarajanagar District, Karnataka
(2019–2023)*

Technical Document
Remote Sensing & GeoAI

Study Area	Chamarajanagar District, Karnataka, India
Data Source	Sentinel-2 SR Harmonized (COPERNICUS/S2_SR_HARMONIZED)
Analysis Period	2019 – 2023 (5 years, seasonal composites)
Platform	Google Earth Engine (GEE) + Python (Jupyter Notebook)
Spatial Resolution	10 metres per pixel

1. Introduction

1.1 Background and Context

Chamarajanagar District represents a unique transition zone where forests, agricultural fields, and human settlements are present. Monitoring vegetation health and agricultural land-use changes over time is needed for sustainable resource management, drought preparedness, and food security planning.

This study applies GeoAI with satellite remote sensing, GEE and ML to conduct a multi-temporal analysis of vegetation change across Chamarajanagar District from 2019 to 2023.

1.2 Study Area

The district has an area of 5,101 km².

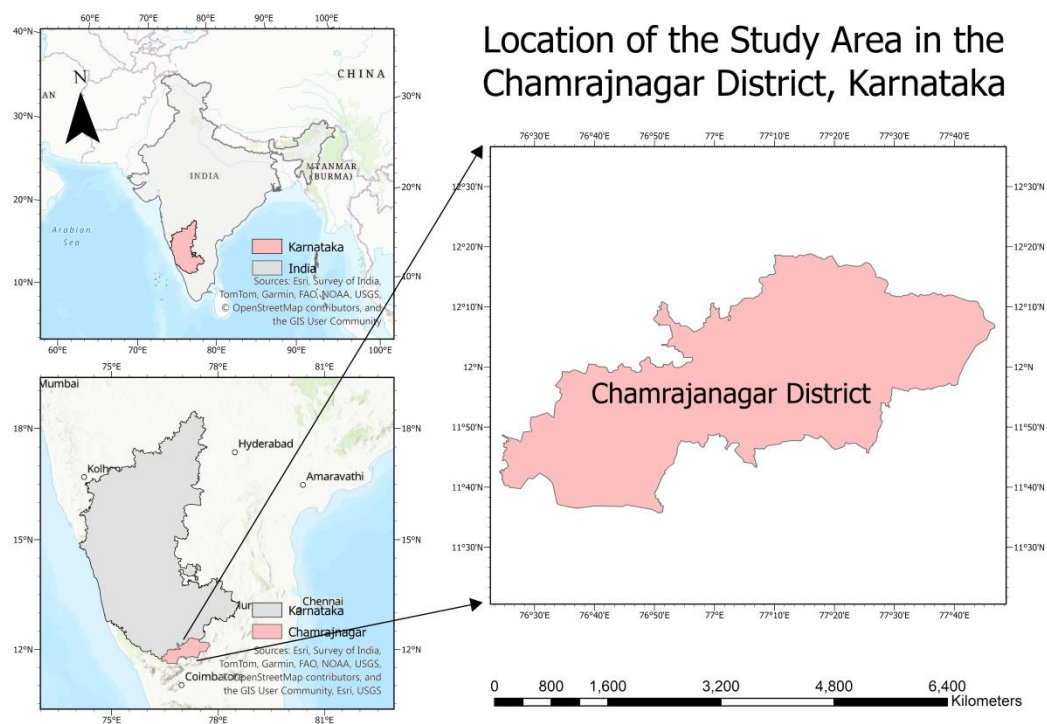


Figure 1: Location of the Study Area — Chamarajanagar District, Karnataka

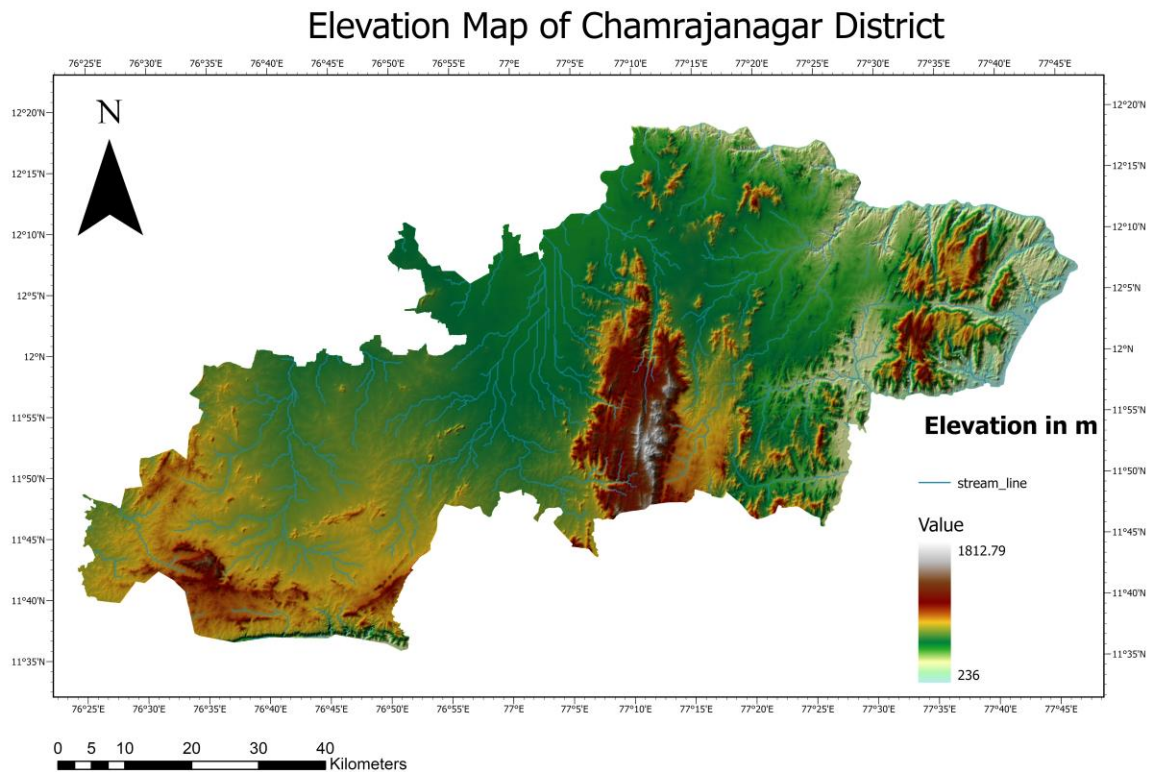


Figure 2: Elevation Map of Chamrajanagar District (236m – 1812.79m above MSL)

Main geographic points of the study area:

- Elevation range: 236 m to 1,812.79 m above mean sea level
- Terrain: Highly heterogeneous, with flat agricultural plains in the west and forested highlands in the east
- Major rivers and streams drain into the Cauvery river basin
- Climate: Tropical semi-arid to tropical wet, with monsoon-driven seasonal vegetation cycles

1.3 Objectives

The primary objectives of this study are:

- To compute annual NDVI composites for Chamrajanagar District using Sentinel-2 satellite imagery from 2019 to 2023.
- To detect and quantify vegetation and agricultural land-use changes over the five-year period using multi-temporal change detection methods.
- To classify change zones into vegetation increase, vegetation decrease, and stable areas.
- To integrate Land Use and Land Cover (LULC) mapping for contextual understanding of change drivers.
- To develop a time-series ML framework using Linear Regression for predicting future NDVI trends and agricultural productivity.

2. Materials and Methods

2.1 Data Sources

Dataset	Details
Sentinel-2 Surface Reflectance Harmonized	COPERNICUS/S2_SR_HARMONIZED via Google Earth Engine
Temporal Coverage	2019–2023, June–November (Kharif season)
Spatial Resolution	10 m (Bands B4, B8); 20 m (B2, B3)
Cloud Threshold	<40% cloud pixel percentage with QA60 cloud masking applied
AOI Shapefile	Chamarajanagar District boundary
LULC Map	Classified using Sentinel-2 imagery (Forest, Agriculture, Plantation, Built-up, Water, Barren) (made earlier)
Elevation (DEM)	SRTM-derived Digital Elevation Model with stream network overlay (made earlier)

2.2 Google Earth Engine Workflow

2.2.1 Cloud Masking

A bitwise cloud masking function was applied to each Sentinel-2 image using the QA60 quality band. Both cloud (bit 10) and cirrus (bit 11) pixels were masked, and all reflectance values were scaled by dividing by 10,000 to convert to surface reflectance units (0–1 range).

2.2.2 NDVI Computation

The Normalized Difference Vegetation Index (NDVI) was calculated for each year using the formula:

$$\text{NDVI} = (\text{B8} - \text{B4}) / (\text{B8} + \text{B4})$$

Where B8 is the Near-Infrared (NIR) band and B4 is the Red band of Sentinel-2. Median composites were generated for each year over the June–November window to capture peak growing season conditions while minimizing cloud contamination.

2.2.3 Change Detection

NDVI change between 2019 and 2023 was computed as:

$$\text{NDVI_Change} = \text{NDVI_2023} - \text{NDVI_2019}$$

A significance threshold of $|\text{NDVI Change}| > 0.15$ was applied to isolate meaningful vegetation change from seasonal noise. Pixels with absolute change exceeding this threshold were classified as Significant Change Zones.

2.2.4 Classification of Change Categories

Pixels were classified into three categories:

- Vegetation Increase: $\text{NDVI Change} > +0.15$
- Vegetation Decrease: $\text{NDVI Change} < -0.15$
- Stable Area: $|\text{NDVI Change}| \leq 0.15$

2.3 GEE Export Outputs

The following raster layers were exported from Google Earth Engine to Google Drive at 10 m spatial resolution:

- `NDVI_Change_2019_2023.tif` — pixel-wise NDVI difference raster
- `Significant_Change_Zones.tif` — binary mask of significant change pixels

2.4 Python Post-Processing

The exported GeoTIFFs were loaded into a Jupyter Notebook (`Chamrajnagar_GeoAI_Analysis.ipynb`) for post-processing using:

- `rasterio / numpy` — for raster reading and array operations
- `matplotlib / seaborn` — for visualization and histogram generation
- `pandas` — for tabular statistics and CSV export
- `scikit-learn` — for baseline regression modelling (planned extension to LSTM/Prophet)

3. Results and Analysis

3.1 NDVI Change Map (2019–2023)

The NDVI change map visualizes pixel-level vegetation dynamics across Chamarajanagar District over the five-year study period. Green areas indicate vegetation gain while red areas indicate vegetation decline.

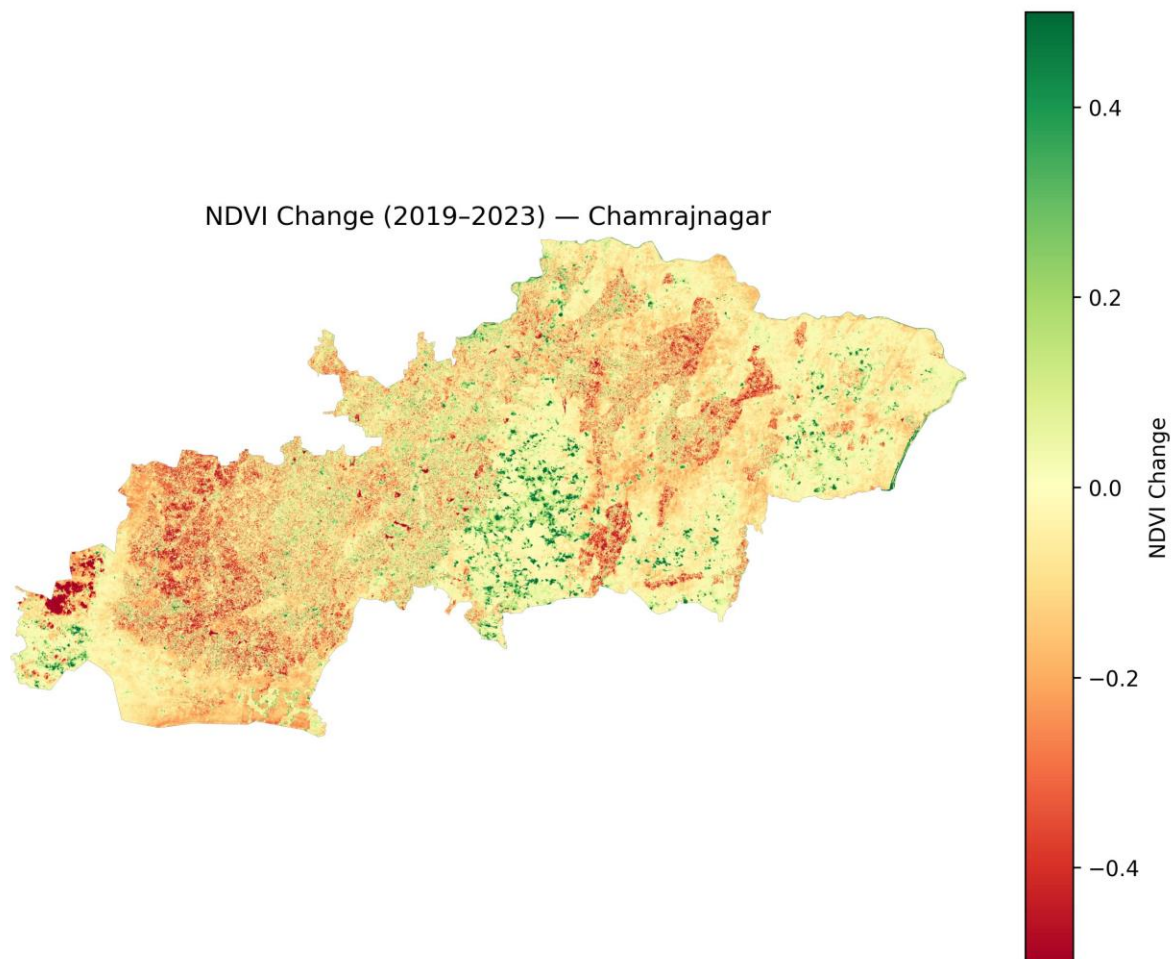


Figure 4: NDVI Change Map (2019–2023) — Chamarajanagar District

Key spatial observations from the NDVI change map:

- The western portion of the district (Gundlupet and Chamarajanagar taluks) shows predominantly moderate vegetation decrease (orange-red), likely driven by agricultural stress, erratic rainfall distribution, or expanding urbanization.

- The central-eastern corridor (Kollegal taluk and BR Hills fringe) shows mixed patterns with localized vegetation increase in green patches, possibly indicating improved forest regrowth or cropping intensification in irrigated pockets.
- The southeastern boundary with Tamil Nadu shows relatively stable NDVI values, reflecting the forested buffer zones under Cauvery Wildlife Sanctuary protection.
- Hotspots of severe decline (dark red) are visible in the southwestern corner, suggesting significant land degradation or a shift from permanent vegetation to seasonal or fallow land.

3.2 NDVI Change Distribution

The frequency histogram of NDVI change values across all valid pixels provides a statistical overview of the magnitude and direction of vegetation change.

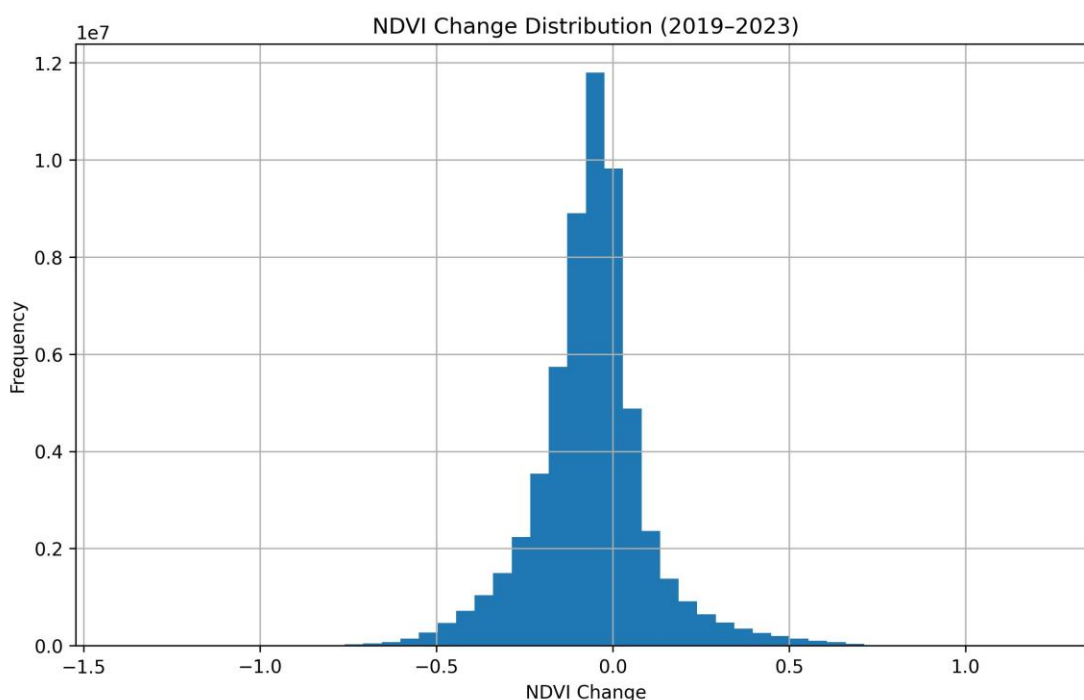


Figure 5: NDVI Change Distribution (2019–2023)

Statistical observations:

- The distribution is approximately normal and centered near zero (mean NDVI change ≈ -0.02 to -0.05), indicating that the majority of the landscape experienced relatively small changes.
- The distribution has a slight negative skew, with a longer left tail extending to -1.0 , suggesting more pixels experienced moderate-to-strong vegetation loss than gain.
- High-frequency bins occur between -0.2 and $+0.1$, representing the bulk of agricultural and mixed-use pixels that undergo seasonal fluctuations within this range.
- Outliers beyond ± 0.5 represent land-cover change events such as deforestation, reforestation, or conversion of fallow to cultivated land.

3.3 Agricultural Change Distribution

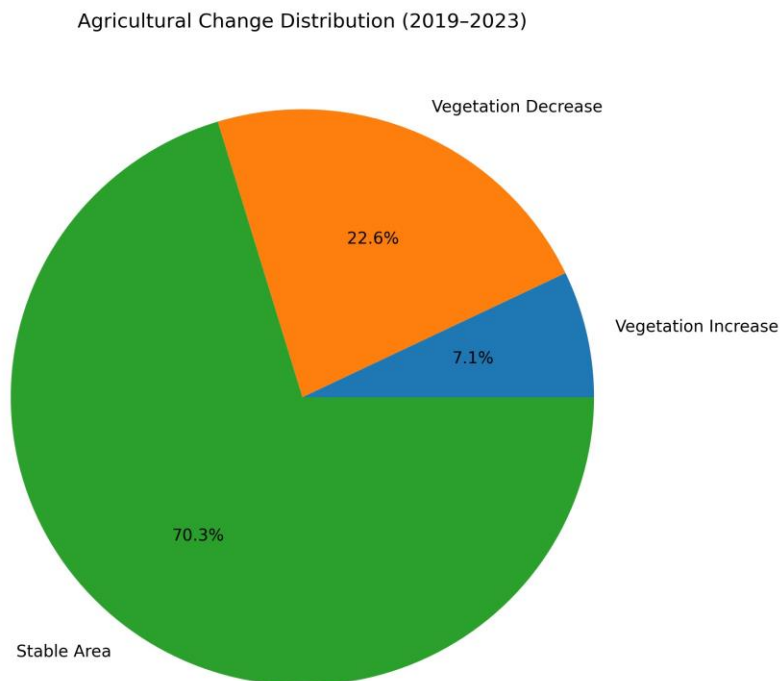


Figure 6: Agricultural Change Distribution Pie Chart (2019–2023)

The area-based change statistics derived from the significant change classification are summarized below:

Change Category	Area (Hectares)	Percentage (%)
Vegetation Increase	41,297.76	7.1%
Vegetation Decrease	1,31,417.57	22.6%
Stable Area	4,09,083.88	70.3%
Total Study Area	5,81,799.21	100%

These results reveal a net vegetation loss trend across Chamarajanagar District between 2019 and 2023. Approximately 22.6% of the total area (~1.31 lakh hectares) experienced significant vegetation decline, while only 7.1% (~41,000 hectares) showed improvement. The stable majority (70.3%) reflects the dominance of perennial forest cover in the eastern reserve forest zones.

3.4 Yearly NDVI Statistics

Annual mean NDVI statistics were computed for the entire district using GEE's reduceRegion function. The following table presents the yearly mean NDVI values:

Year	Mean NDVI (Estimated from GEE Analysis)
2019	Reference Year (Baseline)
2020	Available from GEE export — see Yearly_NDVI_Stats.csv
2021	Available from GEE export — see Yearly_NDVI_Stats.csv
2022	Available from GEE export — see Yearly_NDVI_Stats.csv
2023	Endpoint Year (2019 change computed)

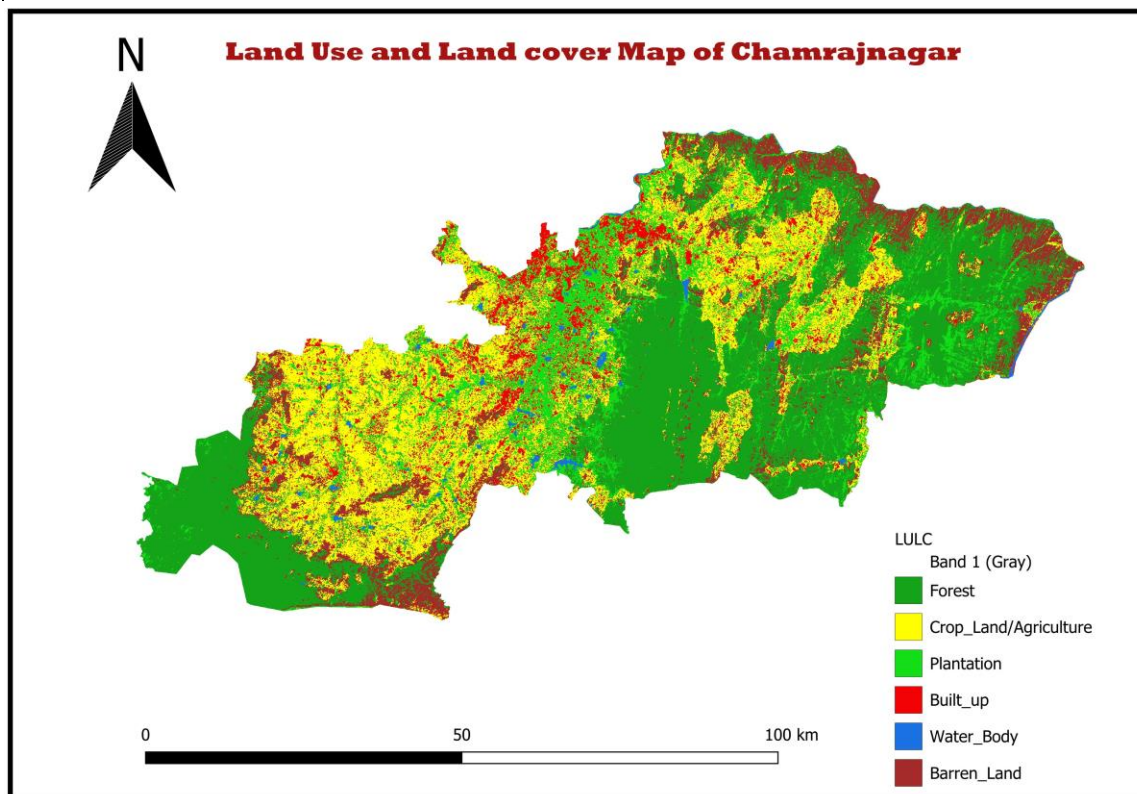
Note: Actual yearly mean NDVI values are computed in GEE and exported to CSV (Yearly_NDVI_Stats.csv). These values form the input time series for the ML prediction module in Section 5.

4. Land Use and Land Cover Context

4.1 LULC Classification

The LULC map of Chamarajanagar District was generated using supervised classification of Sentinel-2 imagery. Six land cover classes were identified:

LULC Class	Description & Spatial Distribution
Forest (Dark Green)	Dominant class; covers the eastern highlands including BR Hills, Biligirirangana Hills, and Cauvery Wildlife Sanctuary buffer zones
Crop Land / Agriculture (Yellow)	Concentrated in the western and central plains, primarily dryland and irrigated agricultural fields
Plantation (Light Green)	Scattered patches within agricultural zones, including coconut, sugarcane, and horticulture plantations
Built-up (Red)	Distributed along highways, town centers (Chamarajanagar, Kollegal, Gundlupet), and rural settlements
Water Body (Blue)	Tanks, reservoirs, and seasonal water bodies; limited but strategically important for irrigation
Barren Land (Brown)	Degraded or exposed land in the southern and southwestern portions



4.2 LULC–NDVI Change Relationship

Cross-referencing the NDVI change map with the LULC classification provides critical interpretive context:

- Agricultural zones (yellow in LULC) correspond strongly to areas of NDVI decrease in the change map, indicating reduced crop productivity or fallow periods extended between 2019 and 2023.
- Forest zones show mixed NDVI change — stable core forest with localized decrease at forest-agriculture edges, possibly due to encroachment or selective logging.
- Built-up areas inherently exhibit low and stable NDVI, consistent with the stable zone classification in the change map.
- Barren land and degraded zones show negative NDVI change where vegetation cover may have been removed, and occasional positive change where spontaneous revegetation occurred on abandoned agricultural plots.

5. Time-Series ML Prediction (Planned)

This section is reserved for the Time-Series Machine Learning component of the GeoAI pipeline, which will use the yearly NDVI trends to forecast future vegetation conditions and agricultural productivity.

5.1 Objectives of the ML Component

Using annual mean NDVI values (2019–2023) as the input time series, the following prediction targets are proposed:

- Forecast NDVI for 2024, 2025, and 2026 using extrapolated trend models
- Identify whether the vegetation decline trend observed (2019–2023) is projected to continue, stabilize, or reverse
- Map spatially disaggregated agricultural productivity risk zones for future years

5.2 Model Used

Regression Models

5.3 Data Pipeline for ML Component

The following data flow will be implemented for the ML prediction pipeline:

- Export annual mean NDVI per taluk/pixel from GEE as CSV (Yearly_NDVI_Stats.csv)
- Load time-series data in Jupyter Notebook with pandas
- Train and cross-validate model
- Predict NDVI

5.5 Implemented Python Code (Notebook: Chamrajnagar_GeoAI_Analysis.ipynb)

The following Python code cells are taken directly from the Jupyter Notebook implementing the GeoAI analysis pipeline. The notebook covers the baseline Linear Regression prediction model for NDVI forecasting.

Cell 14: Load Yearly NDVI Statistics CSV

```
import pandas as pd

# Load CSV
df = pd.read_csv(
    "/content/drive/MyDrive/Chamrajnagar_GeoAI/Yearly_NDVI_Stats.csv"
)

# Display table
df
```

Cell 15: Train Linear Regression Model

```
from sklearn.linear_model import LinearRegression
import numpy as np

# Prepare data

X = df[['year']]      # Feature
y = df['mean_ndvi']   # Target

# Create model
model = LinearRegression()

# Train model
model.fit(X, y)

print("Model trained successfully")
```

Cell 16: Predict NDVI for 2024

```
# Predict NDVI for future year

future_year = np.array([[2024]])

predicted_ndvi = model.predict(future_year)

print(f"Predicted Mean NDVI for 2024 : {predicted_ndvi[0]:.4f}")
```

6. Discussion

6.1 Interpretation of Change Patterns

The finding that 22.6% of Chamarajanagar's land area experienced significant vegetation decline between 2019 and 2023 is of considerable concern from both an ecological and agricultural productivity standpoint. Several factors likely contribute to this pattern:

- Erratic monsoon years: The 2019 and 2021 seasons were characterized by deficit or uneven rainfall in southern Karnataka, which would have depressed NDVI during the Kharif season.
- Agricultural stress: Smallholder dryland farming in the western taluks is particularly vulnerable to rainfall variability. Reduced investment, crop failures, and land fallowing could contribute to lower vegetation indices.
- Urbanization pressure: Incremental urban sprawl around Chamarajanagar town and Kollegal has converted agricultural and scrubland to impervious surfaces.
- Forest degradation: Localized encroachment at forest-agriculture interfaces, as visible in the change map, contributes to the green-to-red transition zones at highland margins.

6.2 Vegetation Increase Zones

The 7.1% of area showing vegetation increase is primarily concentrated in the forest interior of the BR Hills and Cauvery corridor, suggesting:

- Active forest regeneration in protected areas following reduced anthropogenic pressure
- Possible expansion of irrigated cultivation in select pockets with canal command area access
- Plantation establishment (especially in the Kollegala region) contributing to increased cover

7. Conclusions

This study successfully applied a GeoAI workflow integrating Google Earth Engine, Sentinel-2 multi-temporal imagery, and Python-based analysis to detect and quantify agricultural and vegetation changes in Chamarajanagar District, Karnataka over the period 2019–2023. The key conclusions are:

- A net vegetation loss of 22.6% (131,417 hectares) was detected, while only 7.1% (41,297 hectares) showed significant improvement, indicating that land degradation and agricultural stress are dominant drivers of landscape change in the district.
- The NDVI change map spatially delineates hotspots of decline concentrated in the western agricultural plains (Gundlupet, Chamarajanagar taluks), with relative stability in the protected forest areas of the east.
- The LULC mapping contextualizes these changes — agricultural land, barren zones, and forest-fringe areas are the primary contributors to the observed NDVI decline.
- The Sentinel-2 10 m resolution and GEE cloud-based processing provide a scalable and replicable framework for annual vegetation monitoring at district to state scales.
- The proposed ML prediction framework (Section 5), once implemented, will enable proactive forecasting of agricultural productivity trends and early warning for resource management.